MCU interface — C73 P2 plus added CS2

3V3 1
GND 2
SDI/SIO 3
SCLK/SCL 4
CSB 5
D/C 6
RES/RST_N 7
BUSY_N 8
SL_CSB/CS2 9
GND 10
J2
C73_P2_+_CS2_10pin

C73 power distribution and mode straps

L3
3V3 1 2 EPD_3V3
L0603 120R@100MHz

L4
3V3 1 2 EPD_3V3
L0603 120R@100MHz

R10
EPD_3V3 1 2 V_B
1R

C23
V_B 1 2 GND
100uF/6.3V

R11
EPD_3V3 1 2 VDD
0R

R12
EPD_3V3 1 2 VDDIO
0R

C24
EPD_3V3 1 2 GND
0.1uF/25V/0603

R6
GND 1 2 BS0
0R
4-wire SPI mode strap; verify against final C73 ECO

R7
GND 1 2 BS1
0R
4-wire SPI mode strap; verify against final C73 ECO

R_SL
SL_CSB 1 2 VDDIO
47K DNP
Optional CS2 pull-up for reset/default idle

J1 pin4 is no longer C73 VDD; it is SL_CSB routed to J2 pin9.

Panel connector — MF-1750 pin4 is SL_CSB

1 NC
2 TFT_VCOM
3 FPL_VCOM
4 SL_CSB
5 GDRH
6 RESEH
7 GDRL
8 RESEL
9 GDRC
10 RESEC
11 VPC
12 GND
13 PREVGL
14 VPH
15 VSH
16 VSH_LV
17 VSH_LV2
18 VSL
19 VSL_LV
20 VSL_LV2
21 GNDA
22 REFN
23 REFP
24 TSCL
25 TSDA
26 BS0
27 BS1
28 RST_N
29 BUSY_N
30 DC
31 CSB
32 SCL
33 SIO
34 SI1
35 SI2
36 SI3
37 VDDD
38 VDD
39 GND
40 VDDIO
41 VCP2
42 CP2N
43 CP2P
44 VCP1
45 CP1N
46 CP1P
47 CGH1N
48 CGH1P
49 PREVGH
50 VCOMBD
J1
MF-1750_50pin_FPC_adapted from C73

C73 HV boost and negative pump

C2
V_B 1 2 GND
4.7uF/25V/0603
C73 input cap near L1

L1
V_B 1 2 SW_H
10uH 1A

D4
VPH 1 K A 2 SW_H
MBR0530
C73 boost diode

C5
VPH 1 2 GND
10uF/25V/1210

R5
VPH 1 2 PREVGH
0R

C3
SW_H 1 2 NEG_PUMP_H
10uF/25V/1210

D2
PREVGL 1 K A 2 NEG_PUMP_H
MBR0530
C73 negative pump diode

D3
NEG_PUMP_H 1 K A 2 GND
MBR0530
C73 clamp diode

C6
PREVGL 1 2 GND
10uF/25V/1210

Q1
GDRH 4 5 SW_H
3 2 RESEH
SI1308EDL

R4
RESEH 1 2 GND
0.22R 0.2W 1%

R2
GDRH 1 2 GND
1M

C73 original HV topology: only panel connector pin4 is changed.

C73 VPC pump

C1
V_B 1 2 GND
4.7uF/25V/0603
C73 input cap near U2

R1
V_B 1 2 GDRC
1M

Q2
GDRC 4 5 SW_C
3 2 V_B
SI2301

D1
SW_C 1 K A 2 VPC
MBR0530

C4
VPC 1 2 GND
10uF/25V/1210

L2
SW_C 1 2 RESEC
10uH 1A

R3
RESEC 1 2 GND
0.22R 0.2W 1%

Panel-side capacitors and charge-pump pins

C15
TFT_VCOM 1 2 GND
4.7uF/25V/0603

C7
FPL_VCOM 1 2 GND
0.47uF/25V/1210

R8
VDD 1 2 FPL_VCOM
0R

C8
VSH 1 2 GND
10uF/25V/1210

C9
VSH_LV 1 2 GND
10uF/25V/1210

C10
VSH_LV2 1 2 GND
10uF/25V/1210

C17
VSL 1 2 GND
10uF/25V/1210

C18
VSL_LV 1 2 GND
10uF/25V/1210

C19
VSL_LV2 1 2 GND
10uF/25V/1210

C11
REFN 1 2 GND
1uF/25V/0805

C20
REFP 1 2 GND
1uF/25V/0805_NC
C73 driving mark NC

C12
VDDD 1 2 GND
2.2uF/25V/0603

C13
VCP2 1 2 GND
1uF/25V/0805

C21
CP2N 1 2 CP2P
1uF/25V/0805

C14
VCP1 1 2 GND
1uF/25V/0805

C22
CP1N 1 2 CP1P
1uF/25V/0805

C16
VCOMBD 1 2 GND
0.47uF/25V/1210

R9
PREVGH 1 2 VCOMBD
0R
C73 VCOMBD/Vcom option; verify before layout

Bring-up test pads

TP1 3V3
TP2 GND
TP3 EPD_3V3
TP4 V_B
TP5 VDDIO
TP6 VPH
TP7 PREVGH
TP8 PREVGL
TP9 VPC
TP10 CSB
TP11 SL_CSB
TP12 GDRL
TP13 RESEL
TP14 TSCL
TP15 TSDA
TP16 SI1
TP17 SI2
TP18 SI3
TP19 CGH1N
TP20 CGH1P

This is the complete C73-style circuit, not only a 50-pin interposer.

Critical adaptation: C73 original P1 pin4 VDD is replaced by MF-1750 SL_CSB.

C73 P1 pin1 VPP is not connected because MF-1750 pin1 is NC.

Generated in ESP32_C73 workspace
Sheet: /
File: c73_full_mf1750_adapted.kicad_sch
Title: DESPI-C73 Full Circuit Adapted for MF-1750 Pin 4 SL_CSB
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.3 | Id: 1/1